
Rethinking Statistics and Causality: Why Mechanisms Cannot Be Inferred from Data Distributions

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Abstract

Statistical and causal inference have become universal currencies of explanation across the sciences, particularly in domains where underlying mechanisms remain opaque. Their apparent rigor—spanning psychology, economics and biomedicine—rests on the assumption that patterns within data can reveal the processes that generate them. Yet persistent mismatches between empirical predictions and real-world behaviour expose a deeper limitation: mechanisms cannot be inferred from data distributions alone. To address this limitation, we revisit the foundations of both paradigms, showing how statistical inference reduces explanation to geometric alignment, while causal inference, evolved from Bayes’ theorem and graphical models, extends this misstep by conflating probabilistic structure with causal truth. Both expose the same epistemic gap: data encode a lower-dimensional projection of structure, not the mechanism that generates it. We argue that understanding the world follows two routes: one is data-driven, expanding models toward richer function classes to achieve high-precision prediction, as exemplified by modern deep learning; the other is mechanism-driven, proposing and testing structural hypotheses as in the physical sciences. A robust framework requires both: data-driven approaches for high-precision prediction, and mechanism-driven investigation based on hypothesis and experiment.

1 Introduction

Across many scientific domains, statistical and causal inference have become default tools for turning complex observations into interpretable results. Psychology Collaboration [2015], economics Camerer et al. [2016] and biomedicine Ioannidis [2005], in particular, depend on these methods whenever direct access to underlying mechanisms is out of reach. Over time, this dependence has given rise to an illusion of methodological certainty: as long as the analysis is significant, the claim appears sound. Yet the growing number of failed replications—from behavioural studies to clinical trials—suggests that what these methods capture is often a surface regularity rather than a genuine principle of how the world works.

Most researchers acknowledge that correlation does not imply causation, yet the same procedures remain deeply entrenched. The reason is pragmatic rather than epistemic. When mechanisms cannot be measured and sample sizes are limited, statistics offers a way to project uncertainty into mathematical form. Patterns that would otherwise look noisy can be fitted into curves, slopes or conditional dependencies, each assigned a numeric level of confidence. In this sense, statistics does not uncover meaning—it produces geometric regularity in places where the underlying structure remains unknown.

From its very beginning, statistical reasoning has relied on symbols that conceal its geometric nature. Correlation is the cosine of an angle, variance the length of a vector, regression the alignment of a



Figure 1: **Data as lower-dimensional projections of structure.** A single underlying system can give rise to multiple observational projections, each encoding a different low-dimensional representation of the same structure. Once projected and compressed, the original properties become irrecoverable without prior knowledge of the generating process—illustrating why statistical and causal inference cannot recover mechanisms from data distributions alone.

plane Rencher and Schaalje [2008]. These are operations on geometry, not mechanisms. Modern causal inference inherits the same logic, with an added layer of semantic drift Pearl [2009]. Bayes’ theorem—originally a measure-an identity built from the formal definition of conditioning—was later reinterpreted as a calculus of belief through priors and posteriors, and eventually recast in graphical form as if conditional dependence entailed causal direction. Entire causal frameworks were built on this conflation. What unites both traditions is the same epistemic limit: data capture only the projection of a structure, never the process that generates it.

Understanding the world, therefore, follows two complementary yet incomplete paths. One—exemplified by modern machine learning—seeks ever more flexible function approximators to improve prediction, yet remains blind to the mechanisms that generate the data Hornik et al. [1989], LeCun et al. [2015]. The other follows the logic of the physical sciences: starting from structured prior knowledge, it proposes candidate mechanisms and tests whether they can produce the observed phenomena. But this approach falters whenever the underlying structure is too complex or only partially observable Hempel and Oppenheim [1948]. A satisfactory scientific framework must therefore integrate these two modes: models that predict reliably from data and mechanisms that explain how the phenomena are generated.

Our contributions.

- We argue that the replication crisis is a structural consequence of statistical geometry: when projection induces information loss, correct mechanisms can yield misleading statistical relations, and incorrect mechanisms can produce apparently valid ones.
- We re-examine the foundations of statistical and causal inference, showing that correlation, significance, regression, and conditional dependence arose as geometric or algebraic operations on observed data but were later interpreted as explanations of mechanism—an interpretation their mathematics cannot support.
- We distinguish two paradigms for understanding the world: one that improves prediction through increasingly flexible function approximators while remaining mechanistically agnostic, and one that starts from structured prior knowledge to infer candidate mechanisms but falters when those priors fail to capture real complexity.

2 The Root of the Error: Inference from Projected Observations

Modern inference traditions emerged from two pressures:

- **The underlying mechanisms were not specified.** Many fields operated without explicit accounts of how observations are produced. The processes responsible for generating these phenomena remained implicit, ill-defined, or absent from the modeling framework.
- **Conclusions were drawn from minimal observations.** Researchers routinely attempted to make claims about complex systems using small, sparse, or convenience datasets. Limited data were treated as sufficient to characterize the structure of the world.

These pressures encouraged a substitution: data were treated not merely as evidence about the world, but as stand-ins for the processes that produced the observations. Samples were assumed to come from an underlying distribution; the distribution was assumed to reflect stable properties of the world; and stability was taken to imply mechanism. Once this substitution was accepted, inference inverted the logic of scientific explanation. Rather than defining how the world produces observations, researchers attempted to infer mechanisms directly from the observations themselves.

2.1 The Limits of Statistical Inference

Statistical inference operates on projected observations rather than on the underlying system itself. Formally, what is recorded is not the system state $s \in \mathcal{S}$ but a compressed image

$$x = f(s), \quad x \in \mathbb{R}^n,$$

where f discards the structural and semantic content of s . The distribution $P(x)$ therefore captures only the geometry induced by this projection, not the mechanisms that generated the underlying states.

This distinction has a direct consequence for inference. Patterns in x —whatever their form, whether summaries, estimates, or fitted distributions—are properties of the projected space, not of $\mathcal{S}$. They can be estimated with arbitrary precision and still carry no mechanistic content. Precision in the observed space is not evidence of understanding in the underlying one.

2.2 Inference Without Mechanism

The substitution described above does not merely introduce technical limitations—it reframes what inference is taken to accomplish. When data are treated as stand-ins for the processes that produced them, statistical fit becomes a proxy for explanation, and distributional assumptions replace mechanistic ones. The result is a framework oriented toward describing patterns rather than identifying the processes that produce them.

This is not a limitation that additional data or better estimation can overcome. Mechanism is not a property of any distribution over observations; it is a property of how the world responds to intervention. No quantity derived from $P(x)$ —however precisely estimated, however richly structured—can establish what would happen if the underlying system were acted upon. Observational data, by definition, record what occurred under existing conditions. They carry no information about counterfactual states, and it is counterfactual structure that constitutes mechanism.

3 The Illusion of Inference: When Statistics Confuse Geometry with Mechanism

Statistical inference gradually institutionalized a confusion between geometry and mechanism. Fisher, Neyman, and Pearl each increased mathematical rigor, but shifted attention from questions of *how systems generate data* to questions of *how statistics behave under a model* Fisher [1970], Neyman and Pearson [1933], Pearl [2009]. As statistical modeling came to dominate empirical analysis, explanation was increasingly replaced by estimation. The tools grew sharper, but their targets became thinner. We trace this drift through three canonical constructs—correlation, significance, and regression—each representing a historical attempt to extract mechanism from geometry.

3.1 The Geometry of Correlation: What Pearson Actually Measured

Long before computers and data science, researchers sought a numerical way to quantify statistical association. In the late nineteenth century, scientists in biology and social science were collecting

measurements— height, intelligence, income, behavior—without access to the mechanisms that produced them. When causation was unobservable, they turned to co-variation: if two quantities rose and fell together, perhaps they were connected.

KARL PEARSON formalized this intuition by defining a single measure of linear association between two observed variables Pearson [1896]. The resulting coefficient, later known as the Pearson correlation, quantifies how strongly two datasets align in variation rather than in mechanism:

$$r_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{(x - \bar{x}) \cdot (y - \bar{y})}{\|x - \bar{x}\| \|y - \bar{y}\|} = \cos(\theta_{xy}).$$

Formally, this quantity is a normalized inner product between two centered data vectors. Geometrically, r_{xy} equals the cosine of the angle θ between $(x - \bar{x})$ and $(y - \bar{y})$ in a Euclidean data space. Variance corresponds to vector length, covariance to their inner product, and correlation to the cosine of their relative orientation.

Table 1: **Geometric interpretation of the Pearson correlation.**

	Statistical expression	Geometric meaning
Variance (σ_x^2)	$\frac{1}{n} \sum_i (x_i - \bar{x})^2$	Squared vector length of x
Covariance ($\text{cov}(x, y)$)	$\frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y})$	Inner product between x and y
Correlation (r_{xy})	$\frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$	Cosine of angle between x and y

This geometric formulation is elegant yet deceptively simple. Real-world data rarely follow simple linear structures: they arise from complex, high-dimensional processes. Projecting such complexity into a single pairwise angle inevitably discards structural information and can distort apparent relationships.

Table 2: **Approximate angular separation for common correlation values.**

r_{xy}	Cosine interpretation	Angle θ_{xy} (°)
1.0	Perfect alignment	0
0.9	Very strong	26
0.8	Strong	37
0.7	Moderate–strong	46
0.5	Moderate	60
0.3	Weak	73
0.2	Very weak	79
0.0	None	90

As shown in Table 2, even moderate correlation coefficients correspond to surprisingly large angular separations. In psychology, mean reported correlations are close to $r \approx 0.2$, which geometrically corresponds to vectors separated by roughly 79° Richard et al. [2003], Gignac and Szodorai [2016].

Geometrically, this means the variables are nearly orthogonal—barely pointing in the same quadrant. Typical empirical correlations therefore represent only weak geometric alignment. They capture orientation in a vector space, not the mechanisms that generate the data.

To appreciate this limitation, consider the closely related notion of *cosine similarity* Mikolov et al. [2013],

$$\text{sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}.$$

Mathematically, Pearson correlation is simply cosine similarity applied to centered vectors. The distinction lies not in the formula but in the objects being compared. Cosine similarity typically compares representations produced by the same construction procedure, whereas Pearson correlation

relates variables obtained through independent measurement processes. The result is therefore the angle between quantities that may share no underlying semantic structure.

Even when applied to objects of the same representational type, cosine similarity can be misleading: it often captures superficial statistical alignment rather than meaningful structure. If geometric similarity can fail even in this favorable setting, correlation between variables produced by independent measurements is even more fragile. It reproduces the geometry of association, not the mechanisms that generate the data.

Key insight.

Correlation preserves shape, not meaning—and in doing so, mistakes orientation for explanation.

3.2 From Significance to Ritual: The p-Value Illusion

In statistical practice, researchers needed a numerical rule to decide whether an observed pattern looked unusual under a simple baseline. Fisher’s p -value supplied such a rule—a geometric index of deviation rather than a statement about the underlying system Fisher [1970]. Formally, it is a tail probability in the space of a summary statistic:

$$p = P(T(X) \geq t_{\text{obs}} \mid H_0).$$

This quantity lives entirely within a one-dimensional statistic. It reflects the geometry of the null model rather than the structure of the system that produced the data.

The central illusion is straightforward: a tail region in this one-dimensional geometry is treated as evidence about a high-dimensional world. Thresholding converts this geometric position into a semantic claim—“real,” “significant,” “true”—derived from the assumptions of the null distribution rather than from the structure of the underlying system. A one-dimensional statistic cannot reveal the structure of a system regardless of the threshold applied; many distinct mechanisms can produce the same statistic.

This inference relies on the existence of a valid reference distribution. In classical settings this is justified through asymptotic convergence: under i.i.d. observations with finite variance, the central limit theorem guarantees

$$\sqrt{n}(T(X) - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

a mathematical guarantee conditional on imposed assumptions, not an empirical fact about the world. Absent these conditions, tail probabilities cannot be defined, thresholds cannot be calibrated, and the language of significance loses its meaning.

Key insight.

Significance is a one-dimensional shadow mistaken for an explanation.

3.3 Regression as Projection: The Mirage of Explanation

Regression began as a purely geometric tool. In Galton’s original formulation, the fitted line was the orthogonal projection of one centered data vector onto another—a descriptive summary of co-variation rather than a statement about influence Galton [1886], Rencher and Schaalje [2008]. Over the twentieth century, this construct was recast as a mechanistic one: the coefficient

$$\beta = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{(x - \bar{x}) \cdot (y - \bar{y})}{\|x - \bar{x}\|^2}$$

was no longer read as the optimal scalar in a least-squares projection but as the “effect” of x on y . Nothing in its definition refers to mechanisms, transformations, or influence.

This distinction has structural consequences. Two distinct mechanisms can produce identical regression coefficients. The mapping from underlying states $s \in \mathcal{S}$ to observations $x = f(s)$ collapses many possible causal structures into the same projected statistics. Precision in the projected space therefore carries no information about the mechanisms that produce the observed data.

This slippage underlies the widespread misuse of regression as explanation. In psychology, coefficients are treated as cognitive parameters; in economics, as behavioral laws. Models often fit historical data with impressive precision yet fail to generalize, because the geometry of fit contains neither the direction nor the logic of the underlying causal process.

Key insight.

Regression turns geometric projection into causal language.

4 The Illusion of Causality: When Models Confuse Syntax with Semantics

The geometric barrier outlined above makes one conclusion unavoidable: if probability distributions do not encode the structure of the underlying system, no algebra defined on those distributions can recover it. Yet modern statistics, Bayesian reasoning, and causal inference were built on precisely the opposite assumption—that manipulating expressions within a probability space could reveal the processes that gave rise to the data.

The error was not introduced by later misapplication. Conditioning was formalized as a ratio of measures yet treated as an expression of knowledge or relevance Kolmogorov [2018]. Bayes’ identity was an algebraic equality yet reframed as a rule of belief revision de Laplace [1820]. Graphical factorizations were notational devices yet interpreted as diagrams of mechanism. In each case, meaning was imported through linguistic metaphor and intuitive readings of notation—syntax was taken for semantics, and algebra for transformation.

To clarify this structural error, we examine each construct in its original formulation. The problem is not misuse but ontology: these formalisms never contained any account of mechanism. This confusion forms the foundation of the modern causal illusion.

4.1 Conditioning: A Linguistic Metaphor Mistaken for a Mathematical Operation

Conditioning is often treated as a semantically meaningful transformation, as if the notation “ $A \mid B$ ” described a shift of context, information, or assumption. Formally, however, conditioning in probability theory is defined only as a ratio of measures Kolmogorov [2018]:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This expression performs a purely algebraic task: it normalizes the measure of the intersection $A \cap B$ by the total measure of B , producing a dimensionless scalar. The operator “ $\mid$ ” therefore encodes no semantic content; it introduces neither information restriction nor belief refinement. It merely rescales a subset of measure space.

A deeper problem is rarely acknowledged: *probability multiplication can acquire set-theoretic meaning, but probability division cannot*. When two events are independent, the probability of their joint occurrence factorizes:

$$P(A \cap B) = P(A) P(B) \quad (\text{iff independence}).$$

In this special case, scalar multiplication corresponds to a legitimate event-level operation: the intersection of A and B . Outside such structural assumptions, however, the product $P(A)P(B)$ is merely the product of two scalars, with no guaranteed interpretation in the underlying sample space.

In contrast, probability division has no event-level interpretation. The ratio

$$\frac{P(A \cap B)}{P(B)}$$

does not denote an event, a transformation of the sample space, or any operation in measure theory; it merely rescales one scalar by another. Nothing in the mathematics of division guarantees closure within the unit interval—indeed, division can yield values exceeding 1, a clear indication that no underlying event could correspond to the operation. Thus conditioning is not a lawful transformation on events but a notational convention that disguises a purely algebraic normalization.

The confusion persists because the notation “ $A \mid B$ ” mimics the linguistic phrase “given B ,” encouraging readers to treat an algebraic ratio as if it encoded information restriction, relevance, or causal constraint. None of these appear in the definition. Conditioning performs a geometric rescaling in measure space; the semantics are imported by the analyst, not supplied by the mathematics.

Remark: Probability operations rely on artificial assumptions.

In the real world, most probabilities cannot be multiplied, because multiplication requires independence—and independence is rarely satisfied in practice. For example, you cannot meaningfully multiply the probability of a banana with the probability of an airplane. Independence is therefore not a property of the world; it is a constraint imposed to make calculation tractable.

Division creates a deeper problem. Even multiplication is already strange: independence—the condition that makes probability multiplication meaningful—is not a set-level property but an algebraic constraint imposed on the measure. If even multiplication requires such an external assumption, division is worse: it is not a set-level operation and has no event-level interpretation. Many downstream applications—such as conditioning and Bayes’ rule—therefore behave in very strange ways in real-world settings.

4.2 Bayes’ Theorem: An Algebraic Identity Misread as Epistemology

Bayes’ theorem is often introduced as the mathematical foundation of inference, learning, and belief updating. Yet the theorem itself possesses none of these properties. It is nothing more than a symbolic rearrangement of the definition of conditioning:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B)}.$$

This identity introduces no new information, mechanism, or process. It merely restates a relationship among numerical quantities already defined.

Conditioning, however, is not a lawful operation on events. It is only a ratio of scalars with no set-theoretic or mechanistic meaning. Bayes’ theorem, being an algebraic rearrangement of this ratio, cannot acquire epistemic content from an operation that has none.

The first semantic reinterpretation came from statistics. Early Bayesian analyses renamed the three numerical terms in the algebraic identity underlying Bayes’ theorem as “prior,” “likelihood,” and “posterior.” A static equality was presented as an updating procedure, even though nothing in the identity specifies change, learning, or cognition. This interpretation did not arise from the theorem itself. It was created by assigning epistemic roles to the three positions of an algebraic relationship that is, in essence, equivalent to the structure $a \times b = c$. The meaning resides in the naming convention, not in the mathematics.

Later, cognitive science adopted this statistical metaphor and elevated it into a model of reasoning. Hypotheses and evidence replaced sets and events; beliefs and updates replaced ratios and products. But again, none of these concepts appear in the theorem. They are cognitive narratives layered onto an algebraic identity that is indifferent to interpretation.

Bayes’ theorem never contained epistemic meaning. It appeared to gain meaning only because successive communities—first statistics, then cognitive science—misread algebraic syntax as a model of reasoning.

4.3 From Conditional Distribution to Directed Graph: The Syntax Illusion

Once conditioning is recognized as an algebraic ratio rather than a lawful operation on events, the status of graphical models becomes clearer. Bayesian networks were built on this ratio by treating conditional terms as meaningful structural components. Formally, they provide a compact syntactic factorization of a joint distribution Pearl [2009]:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i)).$$

This factorization expresses nothing about mechanism, process, or direction. It is simply one of many ways to rewrite a joint probability as products of scalar ratios.

The illusion arises when this syntactic decomposition is mistaken for structure. A directed edge—introduced as a bookkeeping device—was reinterpreted as a causal arrow, even though conditional terms contain no directional or mechanistic semantics. The expressions

$$P(X | Y) = \frac{P(X \cap Y)}{P(Y)}, \quad P(Y | X) = \frac{P(X \cap Y)}{P(X)},$$

are simply ratios of probabilities. Nothing in the distribution distinguishes $X \rightarrow Y$ from $Y \rightarrow X$, and no factorization order corresponds to a mechanism.

Graphical models therefore encode only syntactic relationships among algebraic ratios. They describe permissible rearrangements of a joint distribution, not the processes that produce it. Interpreting their arrows as mechanisms extends the same category error that treats conditioning as meaningful. It is the syntax illusion: the belief that grammatical form can reveal underlying processes.

4.4 When Syntax Becomes Philosophy: The Expansion of the Causal Illusion

Once conditioning is understood not as a meaningful operation but as a ratio of scalars, and Bayes’ theorem as a restatement of that ratio, the status of causal inference becomes clearer. What later appeared as a philosophical framework began as a sequence of syntactic devices—none of which carried semantics. Directed graphs, conditional terms, and Bayesian factorizations encode permissible algebraic rearrangements, not mechanisms, dependencies, or processes.

Observational data are projections that preserve statistical patterns while discarding the processes that produce them. Once collapsed into data space, no algebraic manipulation—conditioning, factorization, or intervention—can recover what the projection has already erased. The missing structure does not exist anywhere in the observed distribution; it was never recorded.

The field’s history reflects successive attempts to strengthen this substitution. Pearl’s graphical models and *do*-calculus Pearl [2009] treat intervention as algebraic reassignment of conditionals. Rubin’s potential-outcome model Rubin [1974] replaces processes with contrasts between unobservable counterfactual states. Econometric traditions—from Haavelmo to Angrist and Imbens Haavelmo [1943], Angrist and Imbens [1995]—simulate exogenous variation through instruments and natural experiments. All operate solely on observational quantities, performing algebraic rearrangements that cannot, even in principle, recover the processes that produced the data.

Even recent machine-learning approaches, which embed causal reasoning within pattern recognition, inherit the same limitation. By treating structure in data as structure in reality, they instantiate the same categorical mistake across larger observational spaces Athey and Imbens [2016], Johansson et al. [2016], Chernozhukov et al. [2018]. The result is a closed epistemic loop in which causality is defined by syntax rather than semantics, and computation substitutes for understanding.

5 Regrounding Scientific Inference: From Correlation to Mechanism

Scientific inference follows two fundamentally different logics. Both seek structure in data, yet they diverge on what “structure” means.

- **Data-driven prediction.**

Mechanism. The data-driven view treats the world as a mapping problem:

$$y \approx f(x) = \sum_i w_i \phi_i(x).$$

By enlarging the feature space—adding more basis functions ϕ_i —any observed relation can, in principle, be interpolated Hornik et al. [1989]. Deep learning LeCun et al. [2015] pushes this logic to its limit: high-capacity models construct extremely expressive representations that reproduce outcomes across diverse contexts. The model succeeds by matching the geometry of the projected data, not the process that produced it.

Limitation. Once the basis is sufficiently large, infinitely many distinct functions fit the same observations. Because nothing in the mapping identifies which function corresponds to

the underlying process, prediction remains decoupled from explanation. Data-driven models can match every observed case, but they cannot recover—or prefer—the mechanism that generates the world.

- **Mechanism-driven explanation.**

Mechanism. The mechanistic view begins from explicit assumptions about how observations are produced. Mechanisms are first *hypothesized* from prior knowledge and empirical regularities, then tested by deriving the consequences they should generate. This is the epistemology of the physical sciences Hempel and Oppenheim [1948]: mechanisms are proposed, used to derive predictions, and accepted only when experiment confirms that they can generate the phenomenon.

Formally:

$$m \sim \mathcal{M}_{\text{prior}}, \quad x = g(m).$$

Once a component of the mechanism is validated, it becomes a building block: a stable element that can be combined with others to reconstruct increasingly complex processes. Scientific inference proceeds by iterating this loop—observation, hypothesis, derivation, test—allowing mechanisms to accumulate combinatorially until the generative structure of the phenomenon becomes identifiable.

Limitation. Mechanistic reconstruction depends on how much of the true process is observable. When prior knowledge is weak—or when noise, entanglement, or limited access obscures the structure—only fragments of the mechanism can be recovered. Explanation loses resolution even if prediction remains viable. These limits reflect the information the world makes available; they bound the resolution at which mechanism can be inferred.

6 Discussion: The Narrow Window of Statistical Success

The analysis above does not imply that statistical methods never succeed. They succeed only in restricted regimes where the underlying phenomena are sufficiently simple—linear, low-dimensional, and semantically shallow—that little mechanistic information is required for adequate performance. In such settings, the task does not require recovering how the phenomenon unfolds. It requires only a numerically convenient rule that performs acceptably under the observed conditions—predicting values, smoothing noise, or producing stable estimates. When the world is sufficiently simple, coarse geometric approximations can meet these limited demands without recovering the underlying process.

These successes occur because the tasks can be solved without recovering the mechanism—not because inference from projected data is valid. The very existence of these successes produces an illusion of generality: a method that works in one regime appears to validate its assumptions across all regimes. The error arises when such narrow successes are generalized into a universal theory of reasoning. Methods that work only when semantics are minimal have therefore been extended to domains in which semantics constitute the phenomenon—psychology, neuroscience, biomedicine, and economics.

The challenge is therefore not to refine estimation or strengthen asymptotics, but to move beyond geometry-based inference toward intervention and experimentation. Only then can mechanisms be reconstructed rather than approximated from their shadows.

7 Conclusion

In domains where underlying mechanisms are opaque, statistics and causal inference have long served as instruments of certainty. Their appeal lies in translating complexity into geometry—fitting planes, aligning residuals, and manipulating conditional expressions—creating the appearance of explanation even when no mechanism is identified. Yet these operations construct structure within mathematics rather than recover structure from the world; projection can reveal alignment within data but never the process that produced them.

Physical science illustrates the opposite epistemology. Its progress has never come from fitting geometric patterns to observations, but from inferring the mechanisms that make those observations possible. Had physics relied on correlation, regression, or other forms of geometric alignment, it

would have produced an ever-shifting landscape of plausible patterns rather than a stable science of causes. It is mechanism—not geometric fit—that allows explanation to move beyond the surface of data and reach the structure of the world.

Scientific progress ultimately follows only two valid paths. One infers mechanisms from structured priors and observation, as in physics. The other uses high-capacity models to achieve reliable prediction in the projected space without claiming to recover mechanism. Statistical inference offers neither, treating projected data as if it contained the structure that projection has removed. Understanding requires mechanism or prediction—not distributions mistaken for explanation.

Declaration of LLM Usage

The authors used OpenAI’s ChatGPT to assist in refining phrasing and improving clarity. All theoretical arguments and interpretations are original and authored by the researchers.

References

- J. Angrist and G. Imbens. Identification and estimation of local average treatment effects, 1995.
- S. Athey and G. Imbens. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360, 2016.
- C. F. Camerer, A. Dreber, E. Forsell, T.-H. Ho, J. Huber, M. Johannesson, M. Kirchler, J. Almenberg, A. Altmejd, T. Chan, et al. Evaluating replicability of laboratory experiments in economics. *Science*, 351(6280):1433–1436, 2016.
- V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins. Double/debiased machine learning for treatment and structural parameters, 2018.
- O. S. Collaboration. Estimating the reproducibility of psychological science. *Science*, 349(6251):aac4716, 2015.
- P. S. de Laplace. *Théorie analytique des probabilités*, volume 7. Courcier, 1820.
- R. A. Fisher. Statistical methods for research workers. In *Breakthroughs in statistics: Methodology and distribution*, pages 66–70. Springer, 1970.
- F. Galton. Regression towards mediocrity in hereditary stature. *The Journal of the Anthropological Institute of Great Britain and Ireland*, 15:246–263, 1886.
- G. E. Gignac and E. T. Szodorai. Effect size guidelines for individual differences researchers. *Personality and individual differences*, 102:74–78, 2016.
- T. Haavelmo. The statistical implications of a system of simultaneous equations. *Econometrica, Journal of the Econometric Society*, pages 1–12, 1943.
- C. G. Hempel and P. Oppenheim. Studies in the logic of explanation. *Philosophy of science*, 15(2): 135–175, 1948.
- K. Hornik, M. Stinchcombe, and H. White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- J. P. Ioannidis. Why most published research findings are false. *PLoS medicine*, 2(8):e124, 2005.
- F. Johansson, U. Shalit, and D. Sontag. Learning representations for counterfactual inference. In *International conference on machine learning*, pages 3020–3029. PMLR, 2016.
- A. N. Kolmogorov. *Foundations of the theory of probability: Second English Edition*. Courier Dover Publications, 2018.
- Y. LeCun, Y. Bengio, and G. Hinton. Deep learning. *nature*, 521(7553):436–444, 2015.

- T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26, 2013.
- J. Neyman and E. S. Pearson. Ix. on the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, 231(694-706):289–337, 1933.
- J. Pearl. *Causality*. Cambridge university press, 2009.
- K. Pearson. Vii. mathematical contributions to the theory of evolution.—iii. regression, heredity, and panmixia. *Philosophical Transactions of the Royal Society of London. Series A, containing papers of a mathematical or physical character*, (187):253–318, 1896.
- A. C. Rencher and G. B. Schaalje. *Linear models in statistics*. John Wiley & Sons, 2008.
- F. D. Richard, C. F. Bond Jr, and J. J. Stokes-Zoota. One hundred years of social psychology quantitatively described. *Review of general psychology*, 7(4):331–363, 2003.
- D. B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5):688, 1974.